Excretion

15 questions: 5 multiple choice, 10 structured.

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1 · Part 5

Paper 1 & 2 Excretion [1]

Which substance is an excretory product found in human sweat?

A glucose

B starch

C urea

D protein

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2 · Part 5

Paper 1 & 2 Excretion [1]

Which substance is made by photosynthesis and lost from a leaf as a waste product?

A carbon dioxide

B oxygen

C glucose

D water vapour

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3 · Part 5

Paper 2 only Excretion [1]

Where in the human urinary system is urine stored before it leaves the body?

A kidney

B ureter

C bladder

D urethra

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4 · Part 5 Paper 2 only Excretion [1]

Glomerular filtrate passes through several regions of a nephron.

Which region does the filtrate pass through last?

A collecting duct

B loop of Henle

C proximal convoluted tubule

D Bowman's capsule

5 · Part 5 Paper 2 only Excretion [1]

Which substances are all present in the urine of a healthy person?

A water, urea and glucose

B urea, ions and red blood cells

C water, ions and protein

D water, urea and ions

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6 · Part 5

Paper 1 & 2 | Excretion | [5]

Each day, a man loses water from his body in four different ways. The table shows the volume of water he lost in one day by each route.

route of water loss	volume of water lost in one day / cm^3
urine	1350
sweat	450
exhaled air	400
faeces	150

(a) Name the organ of excretion that produces sweat. [1]

(b) Three of the routes in the table involve organs of excretion.

Calculate the percentage of the water lost through organs of excretion that was lost in urine.

Use the formula: percentage = (volume lost in urine ÷ total volume lost through organs of excretion) × 100

Give your answer to one decimal place. [2]

(c) Explain why the removal of faeces from the body is not described as excretion. [2]

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7 · Part 5

Paper 1 & 2 | Excretion | [5]

At night, leaf cells produce carbon dioxide as a waste product of metabolism. This carbon dioxide is lost from the leaf.

(a) Name the process in leaf cells that produces carbon dioxide. [1]

(b) Explain how carbon dioxide moves from a leaf cell to the air outside the leaf. [3]

(c) Give one difference between the way carbon dioxide leaves a leaf and the way carbon dioxide leaves the body of a human. [1]

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8 · Part 5

Paper 1 & 2 Excretion [7]

The lungs are organs of excretion. The table shows the percentage of three gases in the air that Kofi breathes in (inhaled air) and in the air he breathes out (exhaled air).

gas	percentage in inhaled air	percentage in exhaled air
oxygen	21.0	16.0
carbon dioxide	0.04	3.8
water vapour	1.0	6.0

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(a) Calculate how many times greater the percentage of carbon dioxide is in exhaled air than in inhaled air. [2]

(b) Explain why exhaled air contains a higher percentage of carbon dioxide than inhaled air. [3]

(c) When Kofi runs, the volume of carbon dioxide he excretes each minute increases.

Explain why. [2]

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9 · Part 5

Paper 2 only

Excretion

[5]

Each kidney contains about one million nephrons.

(a) The table describes four regions of a nephron. Complete the table by giving the name of each region.

description of region	name of region
a ball of blood capillaries at the start of the nephron	
a cup-shaped structure that surrounds the ball of capillaries	
the first coiled section of tubule, leading away from the cup-shaped structure	
a U-shaped section of tubule between the two coiled sections	

[4]

(b) Name the tube that carries urine from a kidney to the bladder. [1]

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10 · Part 5

Paper 2 only

Excretion

[6]

The table shows the concentrations of four substances in the blood plasma and in the glomerular filtrate of a healthy person.

substance	concentration in blood plasma / g per dm^3	concentration in glomerular filtrate / g per dm^3
glucose	1.0	1.0
urea	0.25	0.25
sodium ions	3.2	3.2
protein	76	0

(a) Describe how glomerular filtrate is formed by ultrafiltration. [3]

(b) Using the table, describe the similarities and differences between the composition of the blood plasma and the composition of the glomerular filtrate. [2]

(c) Explain why there is no protein in the glomerular filtrate. [1]

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11 · Part 5

Paper 2 only

Excretion

[8]

Glucose is filtered from the blood into the glomerular filtrate. In a healthy person, all of this glucose is reabsorbed into the blood in the proximal convoluted tubule. This is called selective reabsorption.

(a) Explain why it is important that glucose is reabsorbed into the blood. [2]

(b) The cells lining the proximal convoluted tubule contain many mitochondria.

Suggest why. [2]

(c) People with untreated diabetes have a very high concentration of glucose in their blood. The table shows the mass of glucose filtered and the mass of glucose reabsorbed each minute by the kidneys of a healthy person and of a person with untreated diabetes.

	healthy person	**person with untreated diabetes**
mass of glucose filtered per minute / mg	125	450
mass of glucose reabsorbed per minute / mg	125	375

(i) Calculate the mass of glucose lost in the urine each minute by the person with untreated diabetes. [1]

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(ii) Calculate the percentage of the filtered glucose that is reabsorbed by the person with untreated diabetes.

Use the formula: percentage = (mass reabsorbed ÷ mass filtered) × 100

Give your answer to one decimal place. [2]

(d) Suggest why the kidneys of the person with untreated diabetes do not reabsorb all of the filtered glucose. [1]

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12 · Part 5

Paper 2 only

Excretion

[7]

Amira drank 1000 cm^3 of water at time 0. She then collected her urine every 30 minutes and worked out the rate of urine production. The table shows her results.

time after drinking / min	rate of urine production / cm^3 per min
0	1.5
30	4.5
60	9.0
90	6.0
120	2.5
150	1.5

(a) Calculate the percentage increase in the rate of urine production between 0 minutes and 60 minutes.

Use the formula: percentage change = ((new value − original value) ÷ original value) × 100 [2]

(b) Explain the change in the rate of urine production between 0 minutes and 60 minutes.

In your answer, refer to the hormone that controls the water content of the blood. [4]

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(c) Suggest why the rate of urine production had returned to 1.5 cm^3 per min by 150 minutes. [1]

13 · Part 5 Paper 2 only Excretion [5]

As fluid passes along the collecting duct, water is reabsorbed into the blood. The fluid surrounding the collecting duct has a higher concentration of ions than the fluid inside the collecting duct.

(a) Explain how water is reabsorbed from the collecting duct into the blood. [3]

(b) The kangaroo rat lives in deserts where there is very little water to drink. Its nephrons have very long loops of Henle. These long loops make the fluid surrounding the collecting ducts even more concentrated than in other mammals.

Suggest how this helps the kangaroo rat to survive in the desert. [2]

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14 · Part 5 Paper 2 only Excretion [6]

The kidneys have two roles: excretion and osmoregulation.

(a) State what is meant by:

(i) excretion [1]

(ii) osmoregulation [1]

(b) In a healthy woman, the concentration of urea in the blood plasma is 0.32 g per dm^3. The concentration of urea in her urine is 20.8 g per dm^3.

Calculate how many times greater the concentration of urea is in her urine than in her blood plasma. [2]

(c) The concentration of urea in the glomerular filtrate is the same as in the blood plasma.

Explain why the concentration of urea in urine is much higher than in the glomerular filtrate. [2]

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15 · Part 5

Paper 2 only

Excretion

[8]

Urea is carried in the blood plasma to the kidneys.

(a) Describe the pathway of a urea molecule from the blood in a kidney until it leaves the body in urine.

In your answer, name the regions of the nephron and the parts of the urinary system that the urea passes through, and the processes that take place along the way. [6]

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(b) The urine of a patient contains protein. A doctor thinks that one region of the patient's nephrons is damaged.

Suggest which region is damaged. Give a reason for your answer. [2]